1. My name is koushik
2. Iam 19 years old
3. The sun rises in the east.
4. I enjoy Playing games
5. AI make our work easier
6. Birds are flying in the sky.
7. We visited exhibition last weekeend.
8. The train arrived at the station on time.
9. A little boy was enjoying stars.
10. Iam studying in KIET.

Tokenized Sentences:

1. ["My", "name", "is", "koushik"]
2. ["I", "am", "19", "years", "old"]
3. ["The", "sun", "rises", "in", "the", "east"]
4. ["I", "enjoy", "playing", "games"]
5. ["AI", "makes", "our", "life", "easier"]
6. ["Birds", "are", "flying", "in", "the", "sky"]
7. ["We", "visited", "the", "exhibition", "last", "weekend"]
8. ["The", "train", "arrived", "at", "the", "station", "on", "time"]
9. ["A", "little", "boy", "was", "enjoying", "the", "stars"]
10. ["I", "am", "studying", "in", "KIET"]

Embeddings Concept

Example 1:

"I enjoy playing games"

Tokens:

["I", "enjoy", "playing", "games"]

Sample Embedding:

Token	Embedding Vector
I	[0.12, 0.45, 0.67]
enjoy	[0.81, 0.34, 0.29]
playing	[0.78, 0.31, 0.41]
games	[0.75, 0.28, 0.39]

The words "playing" and "games" have similar vector values because they are semantically related.

Example 2:

"The sun rises in the east"

Tokens:

["The", "sun", "rises", "in", "the", "east"]

Sample Embedding:

Token	Embedding Vector
sun	[0.92, 0.15, 0.48]
rises	[0.88, 0.20, 0.50]
east	[0.85, 0.18, 0.46]

These vectors help the language model understand that "sun", "rises", "east" are related concepts.

Analysis of Sentences Using RNN and Transformer Approaches

Sentence 1: "My name is koushik".

Tokens:

["My", "name", "is", "koushik"]

RNN Analysis:

- The RNN processes the sentence one token at a time.
- It first reads "My", then updates its hidden state
- Next, it reads "name", using information from "My".
- It continues with "is", remembering the previous words.
- Finally, it processes "Koushik", combining all previous context.

- The final hidden state represents the meaning of the sentence.

<u>Processing Order:</u>

My → name → is → koushik

<u>Transformer</u> <u>Analysis:</u>

- The Transformer processes all tokens simultaneously.
- Through the self-attention mechanism, each words attends to every other word.
- For example:
  - "koushik" attends to "My", "name", and "is"
  - "name" attends to "Koushik" to understand whose name it is
- This allows the model to capture the full context efficiently.

Processing:

[My] ⇔ [name] ⇔ [is] ⇔ [koushik]

(All words interact with each other at the same time).

Sentence 2: "I am 19 years old".

Tokens:

["I", "am", "19", "years", "old"]

RNN Analysis:

- The RNN reads the sentence sequentially
- It processes "I", then "am", then "19", followed by "years", and finally "old"
- The hidden state carries information from previous words to the next word.
- The final hidden state represents the complete sentence.

Processing Order:

I → am → 19 → years → old

Transformer Analysis:

- The Transformer processes all five tokens in parallel.
- Every token can attend to every other token using self-attention.
- For example:
  - "19" connects with "years" and "old" to understand the age.
  - "I" and "am" provide the subject information.
- This enables better understanding of the sentence without relying on sequential processing.

Processing:

[I] ⇔ [am] ⇔ [19] ⇔ [years] ⇔ [old]

(All tokens exchange information simultaneously).

Comparison of outputs RNN and Transformer Models

Sentence 1: "My name is koushik"

- RNN Output
  - Reads the sentence word by word
  - Gradually builds the meaning
  - Correctly identifies "koushik" as the name after processing all words.
  - Accuracy is good for short sentences but depends on sequential memory.

- Transformer Output:
  - Processes all words at the same time.
  - Uses self-attention to directly connect "name" and "koushik"
  - Understands the sentence more efficiently
  - Produces a more accurate contextual representation.

Sentence 2: "I am 19 years old"

- RNN Output
  - Processes the sentence sequentially.
  - Learns the age information step by step.
  - Correctly interprets the sentence but may struggle with longer contexts

- Transformer Output:
    - Analyzes all tokens simultaneously
    - Directly relates "19", "years", "old" using attention
    - Captures the age information more effectively.
    - Provides better contextual understanding.

Accuracy Differences

Processing style:

RNN: Sequential (one word at a time)

Transformer: Parallel (all words together)

Context Understanding:

RNN: Good for short sentences

Transformer: Excellent for both short and long sentences.

Long-Range Dependencies:

RNN: May lose earlier information

Transformer: Maintains relationships between distant words

Speed:

RNN: Slower due to sequential processing

Transformer: Faster due to parallel processing

Memory Handling:

RNN: Hidden state may forget previous information

Transformer: Self-attention preserves important information

Context Accuracy:

RNN: Moderate

Transformer: High

Scalability:

RNN: Less efficient for long text.

Transformer: Highly efficient for long text

Overall Accuracy:

RNN: Good

Transformer: Better than RNN